

# AUTOMATED REVIEW RATING SYSTEM

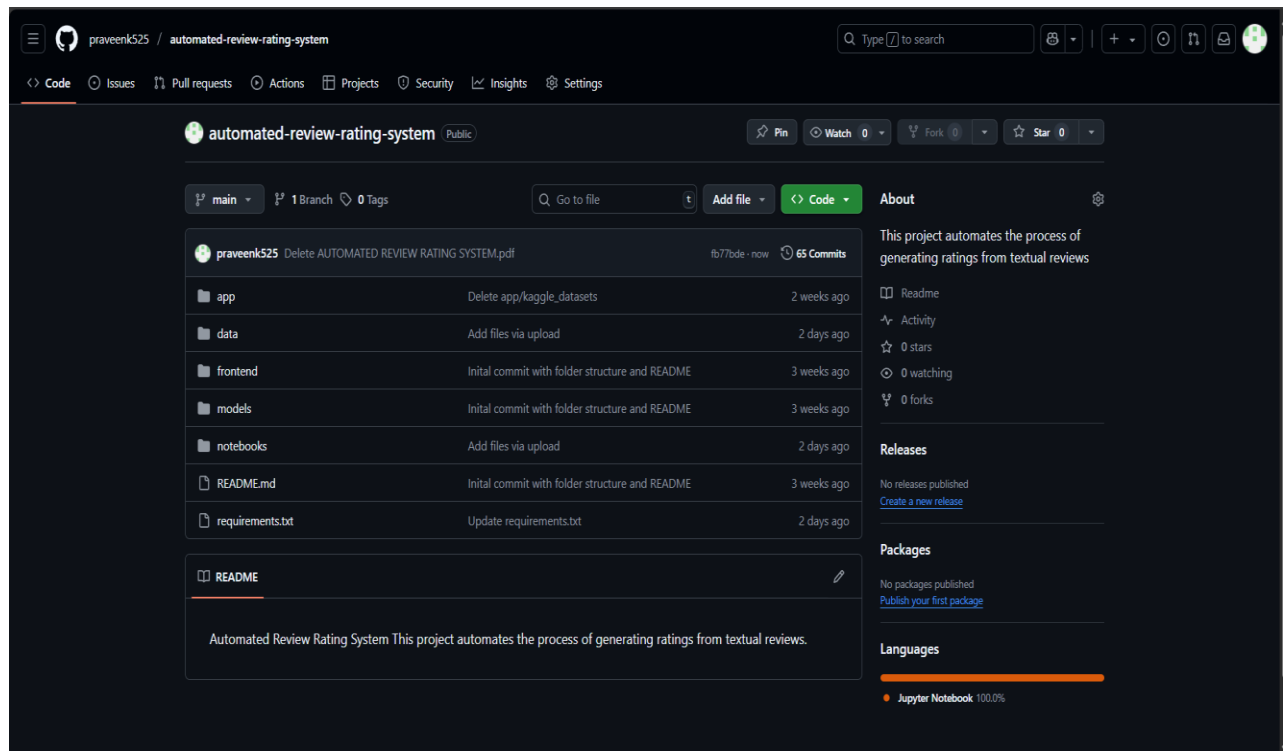
## PROJECT OVERVIEW

This project automates the process of generating ratings from textual reviews. an AI-based system that automatically predicts the numerical rating (e.g., 1 to 5 stars) of a product, service, or business based solely on user-generated textual reviews. This aims to reduce rating manipulation, standardize feedback, and assist platforms in quickly gauging sentiment at scale.

## ENVIRONMENT SETUP

- Installed python with necessary libraries such as numpy, pandas, matplotlib, scikit-learn, seaborn, nltk
- Colab

## GITHUB PROJECT REPOSITORY



## DATA COLLECTION

Data was collected from Kaggle datasets:

For the Automated Review Rating System, a total of 10 datasets were collected, including reviews from Amazon in the Electronics and Movies domains. Each dataset contains two main columns: review text and star rating (1–5).

### 1. Amazon Electronics Reviews

- Contains approximately ~ 6906564 reviews.
- Reviews cover products like phones, laptops, cameras, and other electronics, capturing a wide range of customer sentiments.
- Dataset link: [Amazon-reviews](#)

### 2. Amazon Movies Reviews

- Contains approximately ~3430888 reviews.
- Includes reviews and ratings of movies and DVDs across various genres, reflecting diverse customer opinions.
- Dataset link: [Amazon Movie Reviews Processed](#)

All datasets are merged into one

## DATA PREPROCESSING

Preprocessing is the first and most important step in any machine learning or data science project. It involves cleaning, transforming, and organizing raw data into a format that your machine learning model can understand and learn from effectively.

### Load and Inspect Datasets

- Load all 10 datasets (e.g., Women's Clothing, TripAdvisor, etc.)
- Inspect structure: check column names, data types, sample records
- Verify presence of review and rating columns `pd.read_csv()` → Load CSV files into

a DataFrame `pd.read_excel()` → For Excel files

# View first few rows

`df1.head()`

```
# View last few rows

df1.tail()

# Check column names

df1.columns

# Get DataFrame shape (rows, columns) df1.shape

# Check column data types

df1.dtypes # Full summary

info df1.info()

# Summary statistics (for numeric columns) df1.describe()
```

### **Remove Unnecessary Columns**

- Drop columns not required for prediction, such as:
  - user\_id, product\_id, timestamp, title, image\_url, etc.
- Keep only relevant columns: usually review\_text and rating

### **Remove Null or Missing Values**

- Remove rows with null values in review or rating

```
#Check for missing values in each column

print(df.isnull().sum())

# removing null values

df.dropna(inplace=True)

# Remove rows where review_text or rating is missing df.dropna(subset=['review_text',
'rating'], inplace=True)
```

### **Remove Duplicates and Conflicting Reviews**

- **Exact duplicates:** same review and rating keep one

- **Conflicting reviews:** same review text, different ratings remove all

```
#check duplicates print(df.duplicated().sum()) # Remove all duplicate rows
```

```
df.drop_duplicates(inplace=True)
```

### **Clean Text Data**

- Convert text to lowercase
- Remove:
  - URLs
  - Punctuation
  - Numbers and special characters
  - Extra spaces

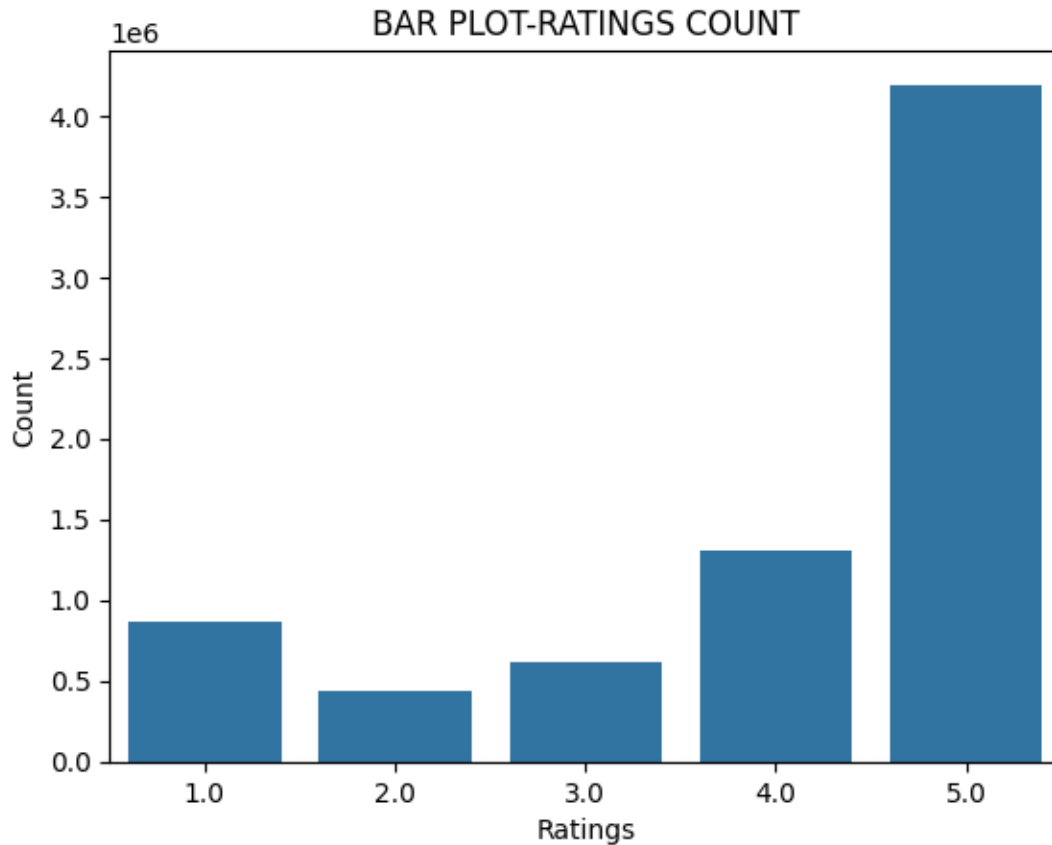
## **DATA VISUALIZATION**

Data Visualization is the process of representing data and information visually using charts, graphs, and other visual elements (like maps or word clouds) to help people understand patterns, trends, and insights in data.

### **BAR CHART**

A bar chart represents categorical data with rectangular bars. The length of each bar is proportional to the value it represents.

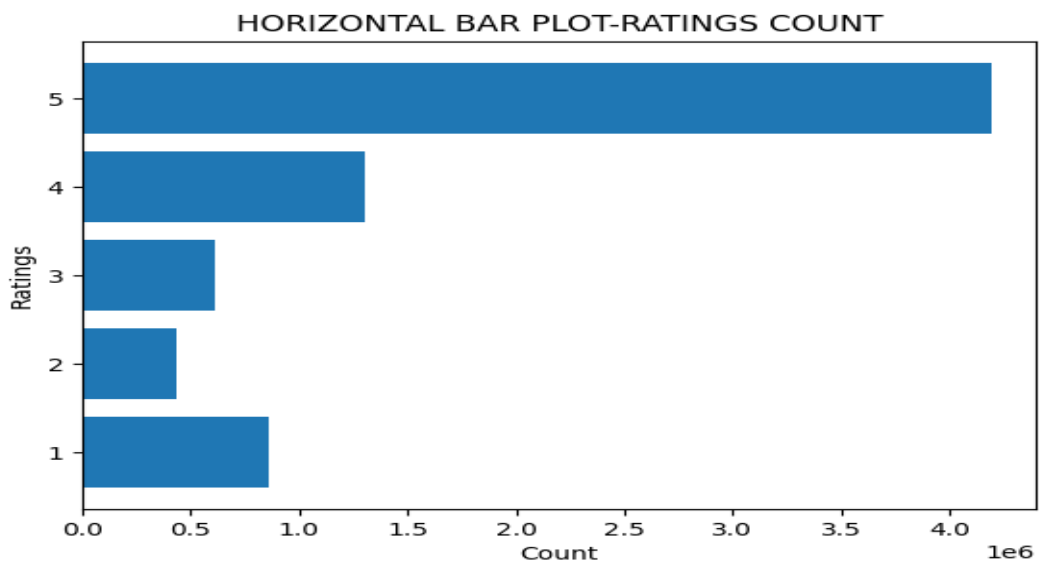
- Use: To show the number of reviews in each star rating (e.g., how many 1-star, 2-star, etc.).
- Number of reviews per rating (1 to 5).



## Horizontal Bar Chart (HBar)

A horizontal bar chart is like a bar chart, but the bars extend horizontally instead of vertically.

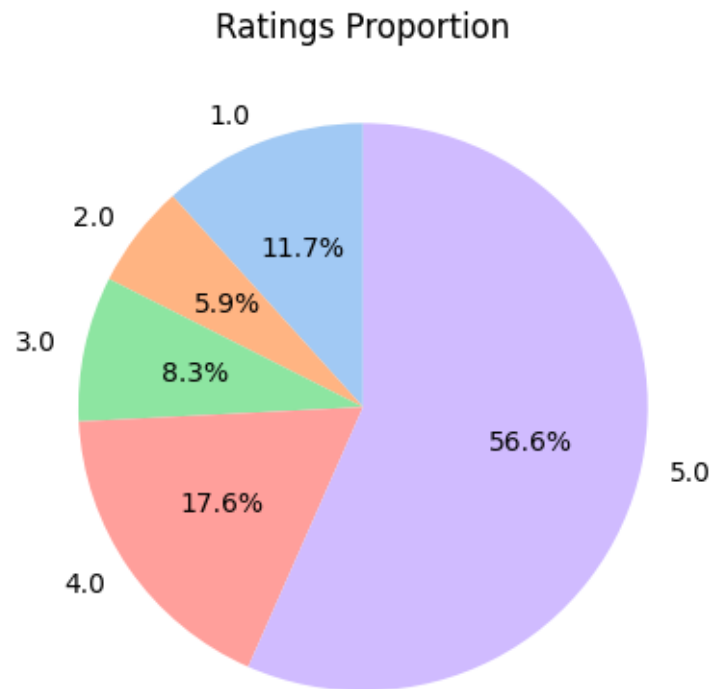
- To show average values like review length for each rating in a clean, readable format.



## Pie Chart

A pie chart is a circular chart divided into slices to illustrate proportions of categories.

- To show the percentage of each rating (how much of the data is 5-star, 4-star, etc.).

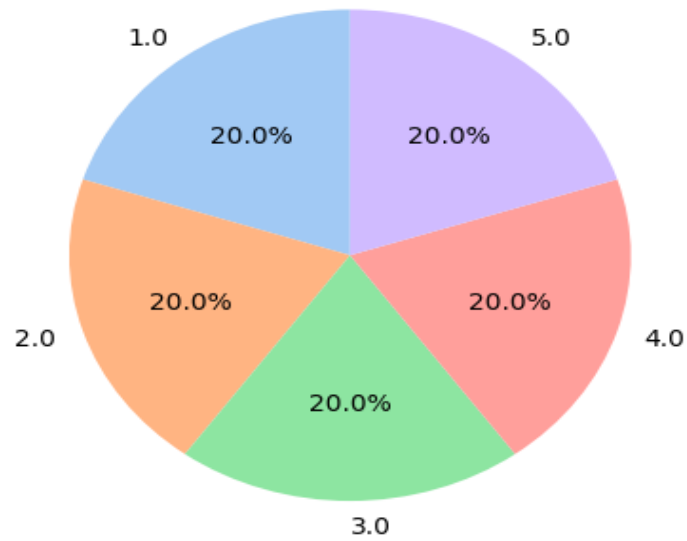


## BALANCED DATASET

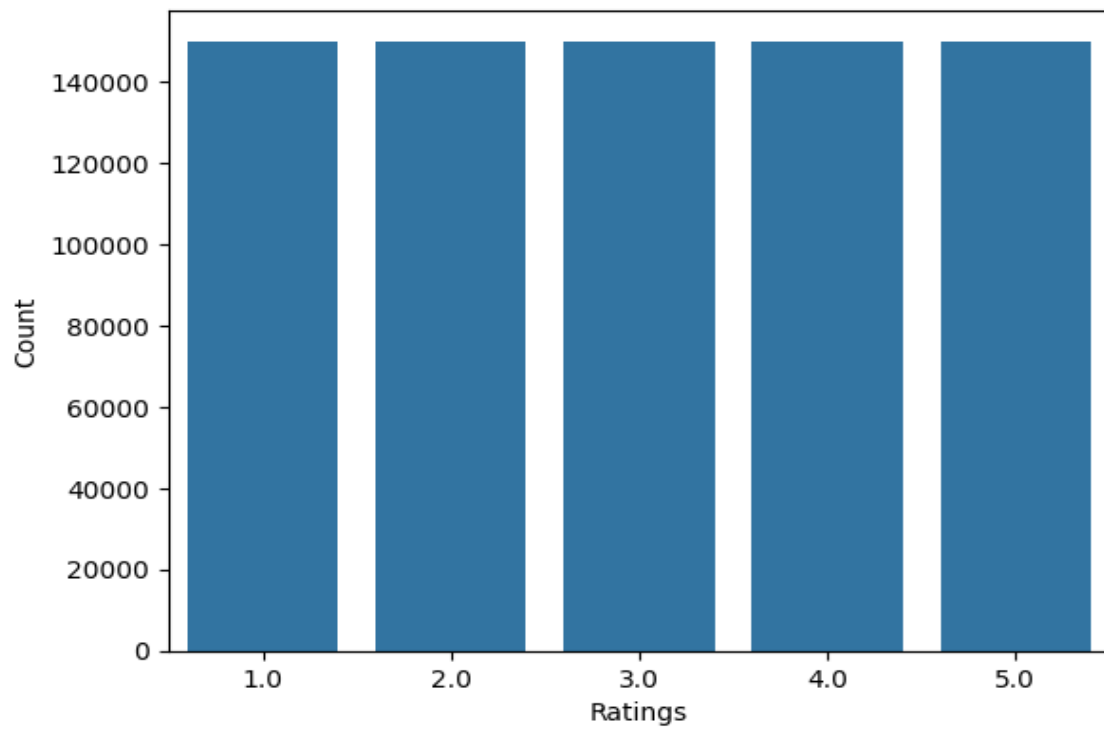
A balanced dataset is one in which all classes or categories have an equal (or nearly equal) number of samples. This ensures that the machine learning model learns from all classes equally and doesn't become biased.

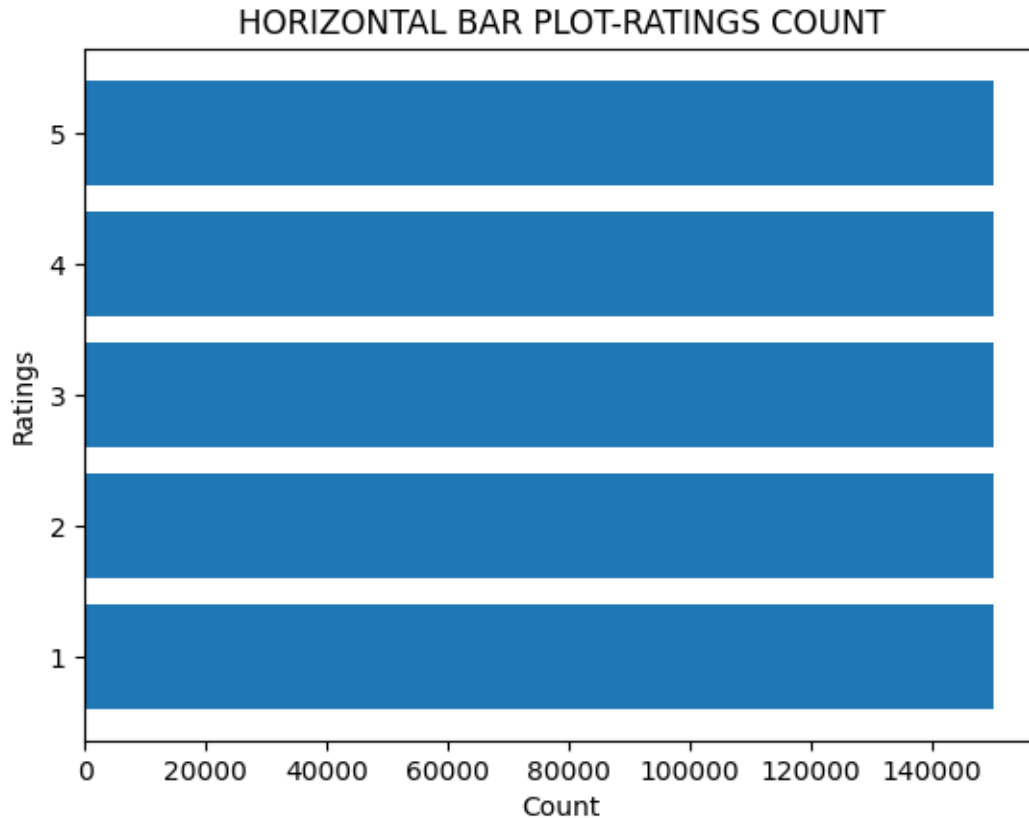
Balanced dataset with 150000 reviews and ratings

Ratings Proportion



BAR PLOT-RATINGS COUNT





## **SAMPLES:**

### **Showing 5 sample review(s) for Rating 1:**

1. lasted hour got very hot then it started to melt downbr stick with oem products from hp i should know better
2. it doesnt fit the nexus nd gen
3. the mouse pointer doesnt track your movement properly you switch it off and then turn it on again to make it work its very frustrating
4. i owned the echo gb which i loved until like hundreds and hundreds of others the screen went dark in under a year livescribes response buy another onebr because i needed this solution to work for aerospace i foolishly bought the hoping id be lucky and it may workbr in the three meetings we tried to use it for it ran very hot omitted much of the recordings and created a low electronic hum for much of the restbr i still have the echo i paid for and because livescribe refuses to even let me pay a fair price to get the display replaced which i shouldnt have to at all as with so many others its a wastebr its sad to see useful technology ruined by poor quality



5. i chose to buy this combo to replace my old logitech solar keyboard which at a couple of years old had developed problems first of all its recharge capacity became severely limited over time to the point where it was almost impossible to charge it in natural light and it was forever giving me low charge warnings secondly a couple of keys on the numerical keypad had become sticky ie they took a while to pop back up after pressing them very annoying as you can imaginebr br i also wanted to replace my logitech mx anywhere mouse which although i loved very much had developed a problem wherein the left mouse button was frequently registering two clicks when i pressed it once again very annoying i noticed that this combo deal included a different mouse the marathon which looked to me to be virtually the same except a little bigger and with a much longer battery lifebr br well this purchase didnt go too well for me first of all the keyboard was shipped with a dime sized patch of moisture under the plastic covering on one of the solar panels i could move it around by pressing on it i thought it might go away in time but after nearly a week it became clear it was there to stay the item was securely wrapped when i got it so i can only presume the moisture got in there at the factory thats some bad quality control it should never have left the premisesbr br onto the mouse i had two problems with this firstly quality control the clickable mouse wheel would sometimes again register two clicks when i pressed it this was inexcusable secondly the design of this mouse compared to the mx anywhere is terrible on the mx anywhere there is a middle mouse button directly under the wheel which by default is set to open a link in a new tab in your browser this is so convenient the marathon mouse also has a middle button but its to toggle the mouse wheel between smooth and notched movement whats more it doesnt work very well i found myself pressing it three or four times to register the change so how do you open a browser link in a new tab with the marathon well you do it by clicking the mouse wheel this sounds ok in theory but in practice its as annoying as hell its almost impossible to click the mouse wheel without actually moving the wheel itself so i frequently found myself scrolling the screen before the click was registered which means you miss the link i cannot describe how infuriating this is compared to the middle mouse button on the mx anywherebr br so the upshot is that i returned this package for a refund i decided instead to purchase the keyboard which i still wanted in a retail store and to purchase another mx anywhere mouse separately it cost about more doing it like this but i figured it was worth itbr br however the new keyboard while being completely free of moisture patches also has a problem the left ctrl key isnt set right and it sometimes sticks when i press it i have concluded that the standard of manufacturing at logitech just aint that hot oh wellmaybe it will unstick in time at this point i cannot be bothered to return yet another itemy well i found myself pressing it three or four times to register the change so how do you open a browser link in a new tab with the marathon well you do it by clicking the mouse wheel this sounds ok in theory but in practice its as annoying as hell its almost impossible to click the mouse wheel without actually moving the wheel itself so i frequently found myself scrolling the screen before the click was registered which means you miss the link i cannot describe how infuriating this is compared to the middle mouse button on the mx anywherebr br so the upshot is that i returned this package for a refund i decided instead to purchase the keyboard which i still wanted in a retail store and to purchase another mx anywhere mouse separately it cost about more doing it like this but i figured it was worth itbr br however the new keyboard while being completely free of moisture patches also has a problem the left ctrl key isnt set right and it sometimes sticks when i press it i have

concluded that the standard of manufacturing at logitech just aint that hot oh wellmaybe it will unstick in time at this point i cannot be bothered to return yet another item

### **Showing 5 sample review(s) for Rating 2:**

1. worked ok but too bulky
2. no instructions and didnt work with my sony nex
3. heres the good and the bad  
pros small size limited power consumption simple setup  
p hdmi output  
cons high price for not including memory ssd or even a power cable  
no usb ports  
intel hd graphics is not compatible with directvpc  
limited storage options  
space  
this little machine is very easy to set up when it comes to adding the ram and msata drive it only takes a couple minutes and its up and running after making a bootable win usb windows was installed within minutes unfortunately win doesnt have appropriate nic drivers for it so those have to be loaded via usb  
the audio and video output is good it handles playback of videos without any issues it works well as a htpc except for one hitch the graphics card will not work with the directvpc application if you want to use that get another machine with a nonintel graphics card  
all and all the price just seems a bit high for what you actually get
4. i was really looking forward to upgrading my original kindle fire i was hoping for a better download experience with this model  
pros the screen resolution is beautiful the sound quality is much improved  
cons it was very difficult getting it hooked up to my home wireless it would not pick up my home network while all other devices in the house had a strong connection my husband who has a masters in computer science after much effort finally got it connected  
i continually loose the connection  
no other devices in my household have this problem i just wonder how im going to connect to other routers when im not at home i probably wont  
i am having a lot a problems downloading games it has a faster download speed than my older kindle but the games are often loaded with glitches  
amazon this is not the way to compete with so many other better performing products out there
5. should have listened to all the people who said the case doesnt stand up like it is supposed to you so much as look at the thing and it falls over cant hold the weight of the tablet on amazon now to buy a different one and will pay attention to the functionality of the stand this time

### **Showing 5 sample review(s) for Rating 3:**

1. this router worked great in the beginning but after months signal strength dropped have to restart often dlink is better over net gear
2. may protect from scratches but picks up fingerprints easily
3. a little loose on the leap but for the price its a good deal pretty colors and very kid friendly
4. i bought two one worked great the other kept popping off the ipad i found that the magnet was installed upside down on the bad one its real easy to just pop the magnet out push something like

a pencil thru the square hole in the back flip the magnet top to bottom and put it back in this should solve the issue great invention but needs a little quality control at assembly

5. this cam works perfect but if you dont know a little about computers like manual installing a device its not for you it does not come with a disk so it installs a generic driver which will not work your cam properly but the cam does have great picture and sound

#### **Showing 5 sample review(s) for Rating 4:**

1. this is a great case and i love the keyboard the selling point for me was that the keyboard had shift buttons i know its a strange reason many keyboards that are available for the ipad are different than normal keyboards but this keyboard is fairly normal also the case is really nice and sturdy but it can be bulky at times i like it being bulky because i often carry around lbs textbooks so it helps to avoid my ipad getting crushed the biggest drawback for me is the severe angle that the ipad sits in when using it in a landscape fashion degree angle overall i recommend this case especially if youre someone who carries big textbooks binders etc

2. cruzer micro is pretty awesome the only problem i have with it is the u which pops up everytime you plug it in however the flash drive works well and looks sleek and with the slider it protects it from breaking

3. item arrived yesterday packaged well cover appears to be well made and adds no real bulk to the mba it does however provide some grip to an otherwise smooth product so far so good

4. the stylus works great the screen cover is nice and the stand feature is a little bit puzzling all in all it was a great bargain

5. the rugrats before the decline as i write this review i cant help but remember when i loved the rugrats it was once my favorite show in television as a kid and when the movie came out in theaters i was more excited than anyone wouldve imagined i was a huge fan the tv show has decreased in quality but the movie still remains solid it introduced a new character to the series and to this day still remains the greatest adventure that the rugrats have gone on didi has recently found out shes pregnant and that means tommy is getting a baby brother when tommys new brother dil is born things around the house change dil begins to receive a bit more attention than tommy and hes always loud and obnoxious the way tommy used to be when he was first born phil lil and chuckie cant stand him and neither can angelica in the meantime stewie has made a new invention a reptar mobile reptar is the rugrats version of godzilla and the babies hero when stu and didi head out for a while they leave the kids home alone with grandpa if youve watched the show you know grandpa is not a good baby sitter he always falls asleep and thats what he does here the babies see this reptar mobile and decide to hop inside and play around angelica the rambunctious three year old brat is trying to watch her show and so she is very annoyed she gives the babies the boot literally and the reptar wagon suddenly slides out the front door tommy has little idea how to control it and neither do his friends phil and lil however dont see it as such a bad thing to them its a chance to take newborn baby dil back to the hospital with the babies gone angelica can finally find peace except theres a problem when she gave the babies the boot they took her favorite doll cynthia with them

now angelica decides she must find them and so she saddles up the family dog spike and spike ends up running out the front door dragging angelica along the result the babies getting lost in the forest far away from home when the parents find out they go on a large search for the babies i suppose what got me to really like this movie was the adventure of it the little imagination of these babies is amazing the movie even begins with a little rendition of indiana jones as theyre actually exploring thier house but when they really arent in a place thats their home you can see just how brave they get never mind how stupid the parents are they never learn from leaving the babies with grandpa the adventure is exciting especially when phil and lil manage to let a bunch of monkeys take dil away soon tommy really is the one responsible for his brother the movie is funny in many places and its songs are enlightening any child will enjoy this movie to death unlike its sequel the babies get into some trouble that isnt too far beyond their control by that i mean the reptar mobile is just a wagon and not some huge gigantic thing destroying the city this adventure is more real than in the sequel rugrats in paris and while dil isnt the most likeable character its unrealistic to think he can do as much as the other babies can he cant walk mind you it would help if dil was as adventurous as his brother but considering hes only a few weeks old we can get by the animation is top notch and very vivid and vibrant much better than the television show ever was its just incredible if there were problems id say its this it is certainly not a movie for grown ups if you grew up with the rugrats as i did you will probably like this movie however grown ups who didnt grow up with the rugrats will be annoyed by their kids loving to watch the film it is clearly aimed at the younger audience its a great movie especially for your kids and especially for nostalgia this was an incredible movie eight years ago and remains a good movie for your kids today

#### **Showing 5 sample review(s) for Rating 5:**

1. this is a handy stylus and pen all in the same utensil it is convenient to have both in one
2. im very happy with this case fits the fire perfectly looks good and was reasonably priced
3. exactly what i needed something send and receive email while traveling
4. great sound quality at a great price exactly what i was looking for in a surround sound system also comes with a easy to use control module very satisfied
5. i love my ipad pink keyboard case the price was very reasonable the keyboard works great i would recommend this product to family and friends

## **Train-Test Split**

### **What is Train-Test Split?**

In machine learning, we split the dataset into:

- **Training Set** → Used to train the model (e.g., 80% of data)
- **Testing Set** → Used to evaluate the model's performance (e.g., 20% of data)

This helps ensure the model can generalize to unseen data.

## Why Use stratify?

If the dataset is imbalanced (e.g., more 5-star reviews than 1-star), a random split could cause:

- Training set to have fewer examples of rare classes.
- Testing set to not represent all classes equally.

Stratification ensures:

- The class distribution (percentage of each rating) is the same in both train and test sets.

## Why Bidirectional LSTM was Chosen

The Bidirectional LSTM (BiLSTM) was chosen because:

- **Captures context in both directions:** Unlike standard LSTM, BiLSTM reads the text sequence forward and backward, which is crucial for understanding sentiment in reviews where meaning depends on surrounding words.
- **Effective for sequential data:** Customer reviews are textual sequences; LSTM handles long-term dependencies better than vanilla RNNs.
- **Proven performance:** BiLSTM models achieve higher accuracy in sentiment and review rating prediction tasks compared to traditional ML algorithms or simple neural networks.

## Preprocessing Steps Used

Before feeding data into the BiLSTM, the following preprocessing steps were applied:

1. **Text Cleaning:** Converted all text to lowercase and removed special characters and punctuation.
2. **Truncating Excessively Long Reviews:** Reviews longer than 300 words were truncated to reduce noise and ensure uniform input size.
3. **Encoding Labels:** Star ratings (1–5) were label-encoded and converted to one-hot vectors for multi-class classification.
4. **Tokenization and Padding:** Text was tokenized using Keras' Tokenizer and sequences were padded to a fixed length (max\_len=200) for uniform input to the model.
5. **Optional Stopword Removal & Lemmatization:** Stopwords were removed and words were lemmatized to reduce vocabulary size and improve semantic understanding.

## Model Training Logic

The BiLSTM model was trained using the following logic:

- **Embedding Layer:** Either random or pretrained embeddings (GloVe) were used to convert words into dense vectors.
- **Bidirectional LSTM Layer:** 128 units with dropout and recurrent dropout to capture forward and backward context.
- **Dropout Layer:** 50% dropout applied after BiLSTM to prevent overfitting.
- **Dense Output Layer:** 5 neurons with softmax activation for predicting one of the five star ratings.
- **Loss Function:** categorical\_crossentropy for multi-class classification.
- **Optimizer:** Adam optimizer for efficient convergence.
- **Callbacks:** EarlyStopping and ReduceLROnPlateau were used to avoid overfitting and adjust learning rate dynamically.

Training was performed using a stratified 80-20 train-test split to ensure proportional representation of all rating classes.

## Evaluation Results and Interpretation

After training, the model was evaluated on the test set:

- **Accuracy:** Measures the proportion of correctly predicted ratings.
- **Precision, Recall, F1-Score:** Evaluated per class to see how well the model predicts each rating.
- **Confusion Matrix:** Visualized misclassifications between ratings.

Accuracy: 0.5498  
Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 1            | 0.66      | 0.62   | 0.64     | 30000   |
| 2            | 0.45      | 0.52   | 0.48     | 30000   |
| 3            | 0.46      | 0.48   | 0.47     | 30000   |
| 4            | 0.52      | 0.48   | 0.50     | 30000   |
| 5            | 0.71      | 0.64   | 0.67     | 30000   |
| accuracy     |           |        | 0.55     | 150000  |
| macro avg    | 0.56      | 0.55   | 0.55     | 150000  |
| weighted avg | 0.56      | 0.55   | 0.55     | 150000  |

